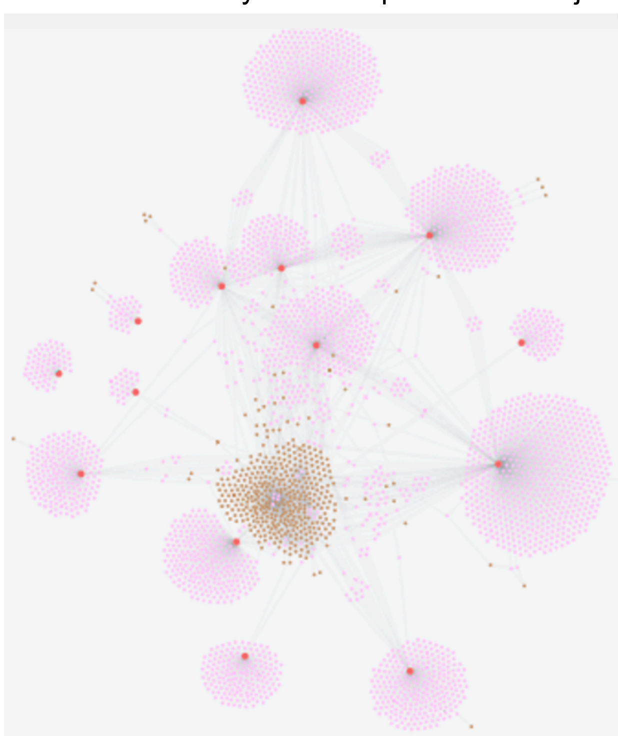
Will Young
Student ID: 924230057
Week 13 Report
4/23/2026

Hello Professor,

Below is an update for my week 13 progress. In addition, I have pushed my updates to my github repo: https://github.com/Woyoung21/LastMile-Sec. For week 13, the main task was to begin writing my paper. However, since I had some extra time in my timeline, I decided to work on my web interface some more and try to retrain my Mistral (Mapper) model. I will discuss both parts in detail and follow up with next week's tasks as well.

As a reminder, last week, I created the web interface for this project. Up until then, everything was backend. And qualifying for the SFSU student AI awards made me develop these visuals a bit more than I originally intended. I mainly did this because I have to submit a 5 minute pitch video on the 27th, and showing my product is much better than just talking about my product. At the end of last week, 3 of the 4 main pages were finished, the Upload page, MITRE Map page and the Remediation Center. The last page that I was unsure about was the Correlation page. After thinking about what we could use as a visual for that page, I decided to try and replicate what my Neo4J RAG database looks like. Since that section does RAG correlation, seeing all the vendor nodes with links to their controls would help me tell the story that our product is grounded in documents that represent industry best practices for hardening their hardware/software. In the Neo4J browser, here is what my graph database looks like. The orange nodes are the vendors, the pink nodes are the controls, and the brown nodes are the MITRE IDs. And when we are using our RAG correlation, these are the actual nodes that are being searched for, each one representing a CIS or NIST control for a specific vendor. Seeing the data can help someone understand why that is important versus just making a statement.



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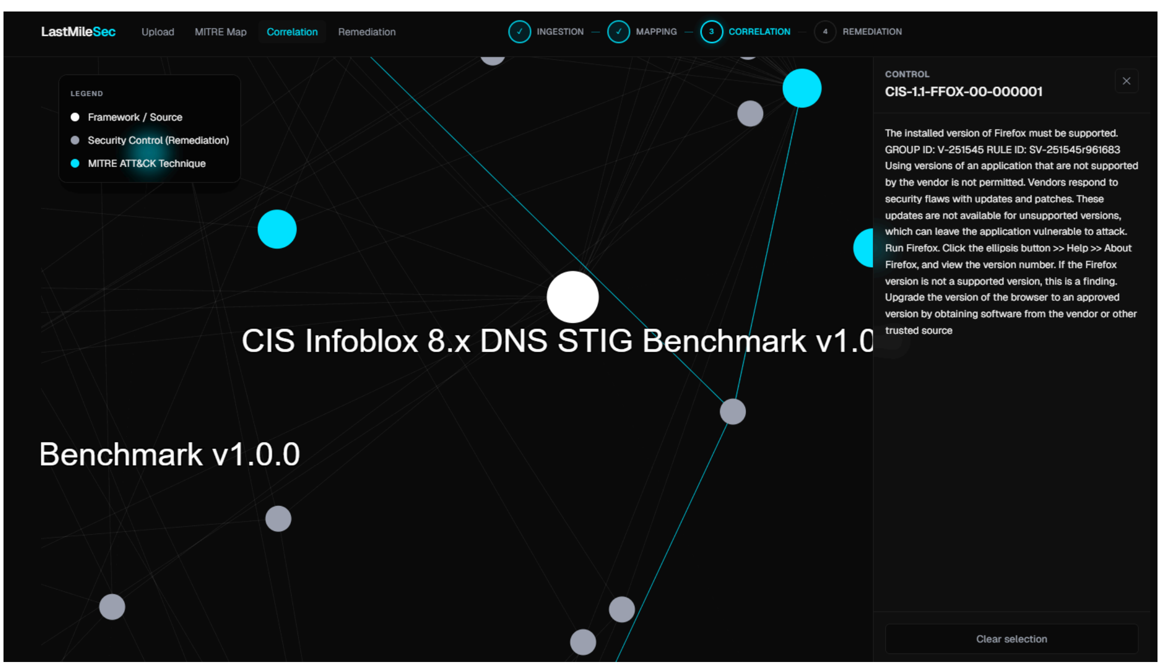
So I wanted to have a page that highlights that very important part of my project. In this screen, all the Cyan nodes are MITRE ID/Techniques, the purple nodes are the vendors (macSO, Windows Server, NIST etc) and the gray nodes are the controls. While this is a screenshot, this page is dynamic, and utilizes a physics engine that allows you to move the vendor nodes, which in turn pulls the associated control nodes, and then affects the surrounding nodes as well.

You can also click onto any node, and a “camera zoom” action happens which zooms in on that node, and an information card on the right side of the screen pops up, displaying the information of each node.



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I thought this was another important feature to “show” the data as it helps reinforce that every single node in this image represents a very important piece of data that can be used to help us develop those customized remediation rules for our lower level technicians. These visuals will help tell the story of our very technical backend to a non-technical audience. I should also mention that these nodes are built from the actual corpus I populated our RAG with.

In addition to creating this visualization, I started retraining my Mistral model. As I began testing the end to end pipeline, I began to notice a MITRE ID clustering. The model seemed to be collapsing onto just a few answers, sometimes mislabeling events all together. I will note that the remediation recommendation for those events were accurate, as we also feed in our technical summary to the Correlation engine and Remediation engine. So the added context helped cover us from potentially mislabeled IDs. Regardless of that, I wanted to try and retrain my Mistral model, to both

Using EarlyStoppingCallback without load_best_mod

[1800/2400

Step	Training Loss	Validation Loss
200	0.808961	0.722666
400	0.780528	0.685302
600	0.702330	0.670527
800	0.660632	0.652967
1000	0.560284	0.659142
1200	0.570971	0.643763
1400	0.562394	0.629498
1600	0.416177	0.655015
1800	0.428803	0.644859

Training complete! RAG-aware adapter saved to: /c

The tokenizer has new PAD/BOS/EOS token

Step	Training Loss	Validation Loss
200	0.357522	0.386486
400	0.230636	0.305223
600	0.151795	0.273712
800	0.111348	0.237994
1000	0.085693	0.227885
1200	0.075722	0.216383
1400	0.054896	0.221395
1600	0.052343	0.208797
1800	0.051341	0.208225
2000	0.042726	0.204956
2200	0.038370	0.204689
2400	0.039870	0.205090

From the beginning, you can see there is a large difference. The most recent notebook sanitized the dataset we used, rather than relying on the existing dataset from huggingface. This cleaner, more structured data allowed us to start at a lower loss percentage. Additionally, I ensured the model was trained on all IDs, and even made more rare IDs show up multiple times in the training data. The goal was to enlarge the model's ability to recognize different IDs based on a text summary, instead of collapsing to its favorite ID. Below is a chart that will show the head to head comparison of the two models so far. I will try to revise some more to increase our accuracy, but this is a great start. I will also mention, all in this fine-tuning session took a little over 6 hours, even with a high-RAM A1000 GPU and cost about 60 units of compute.

Metric	V2 Performance	V3 Performance	Improvement
Micro-F1 (Accuracy)	0.1720	0.2045	+18.9%
Macro-F1 (Diversity)	0.0223	0.0493	+121.1%
Top-1 Fraction (Bias)	0.1483	0.0398	-73.2% Reduction
Final Val Loss	~0.64	~0.20	-68.7% Reduction

- **Micro-F1:** This measures the model's total accuracy across all predictions; our **18.9% increase** proves that the V3 adapter is fundamentally better at getting the right answer more often.

- **Macro-F1:** This evaluates how well the model handles rare vs. common categories; **more than doubling** this score confirms the model successfully learned the "long tail" of complex MITRE techniques instead of just the easy ones.

- **Top-1 Fraction:** This tracks "repetition bias" or the tendency of the model to guess its favorite tag when it's confused; our **73% reduction** shows the model has stopped hallucinating common tags and is now making intentional, data-driven decisions.

- **Final Val Loss:** This represents the mathematical error during training; a **68% lower loss** in V3 indicates a much more stable and "smarter" model that has truly internalized the relationship between technical summaries and attack IDs.

Overall, I am happy with the improvement and will continue to test over the weekend. I may modify the training notebook more to see if we can continue to improve this model. I will also be working on my 5 minute pitch for the student competition. Next week I will begin my final paper as well.